

Text to Haptics: Method and Case Studies of Designing Tactile Graphics for Inclusive Tactile Picture Books by Digital Fabrication and Generative AI

Kengo Tanaka
University of Tsukuba
Tsukuba, Japan

kengo.tanaka@digitalnature.slis.tsukuba.ac.jp

Ayaka Tsutsui
University of Tsukuba
Tsukuba, Japan

ayakatsutsui1128@digitalnature.slis.tsukuba.ac.jp

Tatsuki Fushimi
University of Tsukuba
Tsukuba, Japan

tfushimi@digitalnature.slis.tsukuba.ac.jp

Yoichi Ochiai
University of Tsukuba
Tsukuba, Japan
wizard@slis.tsukuba.ac.jp



Figure 1: Creating a tactile picture book using a 3D printer.

ABSTRACT

In this case study, we explore the possibilities between Generative AI and tactile graphics for inclusivity in computer graphics communities. The use of Generative AI in the design of tactile graphics has made it possible to support the processes used by publishers and tactile graphics designers. In addition, the idea of printing tactile graphics on transparent sheets with a 3D printer through digital fabrication technology allows the creation of inclusive tactile picture books that can be read and enjoyed together by sighted and visually impaired people in a single picture book.

CCS CONCEPTS

• **Human-centered computing** → **Interface design prototyping**.

KEYWORDS

tactile graphics, tactile picture books, generative ai, 3D printing, visual impairments

ACM Reference Format:

Kengo Tanaka, Tatsuki Fushimi, Ayaka Tsutsui, and Yoichi Ochiai. 2023. Text to Haptics: Method and Case Studies of Designing Tactile Graphics for Inclusive Tactile Picture Books by Digital Fabrication and Generative AI. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH '23 Labs)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588029.3595471>

1 INTRODUCTION

There are two types of tactile picture books: tactile picture books printed with ultraviolet (UV) curing resin ink sold by publishers and tactile picture books made by volunteers who cut and paste transparent sticker-like sheets by hand. However, these tactile picture books have several problems. Commercially available tactile picture books are limited in variety and are more expensive than picture books. Tactile picture books handmade by volunteers cannot be produced in large quantities because they take a long time to produce and are not suitable for creating detailed tactile graphics. In addition to commercially available tactile picture books and handmade tactile picture books, there have been previous studies of tactile picture books produced by 3D printers [Kim et al. 2014; Kim and Yeh 2015; Stangl et al. 2015, 2014] and laser cutters [Johnson et al. 2022]. However, these studies are designed on the assumption that only visually impaired people can enjoy tactile picture books, and therefore, sighted people cannot confirm the original illustrations of the picture books. In this paper, we examine a design process that enables even beginners to create tactile graphics by utilizing img2img of Generative AI, and propose a design method

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Labs, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0153-5/23/08.

<https://doi.org/10.1145/3588029.3595471>

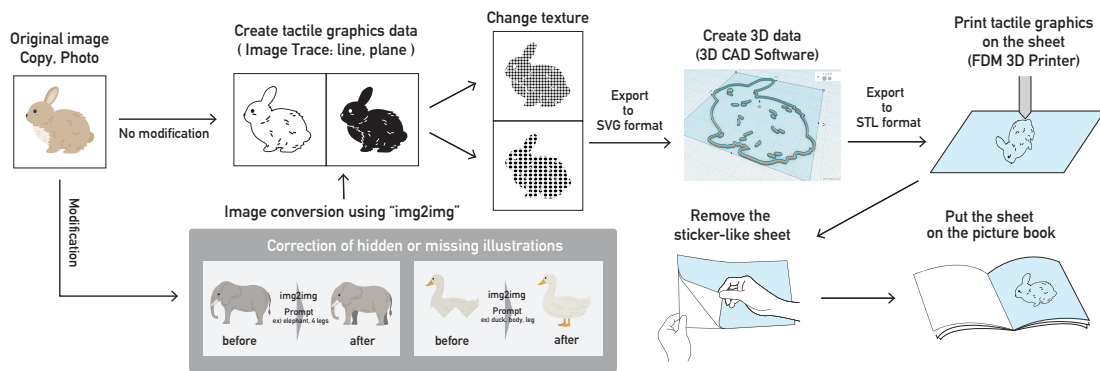


Figure 2: Overview of the design system for a tactile picture book

for tactile picture books that can retain the original illustrations of the picture book while using a 3D printer. This will make it possible to create a picture book that can be read and enjoyed together by both sighted and visually impaired people.

2 DESIGN AND IMPLEMENTATION

We propose a design method to transform an ordinary commercial picture book into a tactile picture book by creating tactile graphics using Generative AI and printing the tactile graphics on a picture book using a 3D printer, as shown in Figure 2.

The design of tactile graphics does not follow the original illustration of the picture book, and it is necessary to modify the illustration depending on the situation. In this case, we believe that the img2img function of Generative AI can be used to lower the technical hurdle in designing tactile graphics. When an illustration of a character is missing or hidden by objects or depth, the entire body of the original character must be restored. These restored images are then used to create path data from the illustration image using the image tracing function of image editing software, and 3D CAD software is used to create the 3D data for the tactile graphics. The 3D data of the tactile graphics is printed on a transparent sheet using Fused Deposition Modeling (FDM) 3D printer. If we try to print directly on a picture book, we will need to adjust the printing position, but by printing on a transparent sheet, we can adjust the position by ourselves when attaching it to the picture book. By using transparent Thermoplastic Polyurethane (TPU) filament in this process, we can create a tactile picture book that is difficult to peel off even when the paper is folded. In addition, because the transparent material does not hide the illustrations when pasted over the illustrations in the picture book, visually impaired and sighted people can enjoy the picture book together.

3 THE SIGGRAPH ATTENDEE EXPERIENCE

3.1 Demo

Attendees will be able to touch the tactile picture books we have created to understand them, and they will also be able to experience creating their own tactile picture books. We plan to set up a space where attendees can 3D print their own designs based on the library and tactile graphics prepared in advance. Craft materials needed

to create tactile picture books, such as transparency sheets and filaments, will also be provided. Attendees will be able to take home the picture books they create at the demo booth.

3.2 Designing Inclusive Tactile Picture Books

The design of tactile graphics is not necessarily a one-to-one correspondence with the original illustration. Different tactile graphics need to be designed for different ages and different senses. Through this demonstration, you can create an original tactile picture book personalized for attendees. Our method can be inexpensively created by combining commercially available products and can be imitated at home. This demo will make people aware of the existence of tactile picture books and create opportunities for many people to enjoy picture books by creating inclusive picture books.

ACKNOWLEDGMENTS

This work was supported by JST, CREST Grant Number JPMJCR19F2, Japan.

REFERENCES

- Nicole Johnson, Tom Yeh, and Ann Cunningham. 2022. Low-Cost Tactile Coloring Page Fabrication on a Cutting Machine: Assembly and User Experiences of Cardstock-Layered Tangible Pictures. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (Athens, Greece) (ASSETS '22)*. Association for Computing Machinery, New York, NY, USA, Article 31, 5 pages. <https://doi.org/10.1145/3517428.3551353>
- Jeeun Kim, Abigale Stangl, Ann Eisenberg, and Tom Yeh. 2014. Evaluating Tactile User Experience with Tactile Picture Books for Children with Visual Impairment. *Tactile Evaluation Methods Workshop* (2014).
- Jeeun Kim and Tom Yeh. 2015. Toward 3D-Printed Movable Tactile Pictures for Children with Visual Impairments. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (Seoul, Republic of Korea) (CHI '15)*. Association for Computing Machinery, New York, NY, USA, 2815–2824. <https://doi.org/10.1145/2702123.2702144>
- Abigale Stangl, Chia-Lo Hsu, and Tom Yeh. 2015. Transcribing Across the Senses: Community Efforts to Create 3D Printable Accessible Tactile Pictures for Young Children with Visual Impairments. In *Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility (Lisbon, Portugal) (ASSETS '15)*. Association for Computing Machinery, New York, NY, USA, 127–137. <https://doi.org/10.1145/2700648.2809854>
- Abigale Stangl, Jeeun Kim, and Tom Yeh. 2014. 3D Printed Tactile Picture Books for Children with Visual Impairments: A Design Probe. In *Proceedings of the 2014 Conference on Interaction Design and Children (Aarhus, Denmark) (IDC '14)*. Association for Computing Machinery, New York, NY, USA, 321–324. <https://doi.org/10.1145/2593968.2610482>